

Comprehensive Bulk RNA-seq Teaching Guide

This guide compiles foundational RNA-seq datasets and advanced host-pathogen interaction studies to provide a complete pedagogical framework for teaching bulk transcriptomics. By traversing from canonical simple-design experiments to complex mixed-species models, students gain exposure to the full spectrum of modern bioinformatics challenges.

Part 1: Foundational Bulk RNA-seq Studies

These studies represent the gold standards for introducing basic bulk RNA-seq concepts. They are widely used as the benchmark datasets by the authors of major analysis packages (e.g., DESeq2, edgeR).

1. The "Airway" Study: Human Smooth Muscle Cells (Himes et al., 2014)

Context: Profiling airway smooth muscle cells treated with dexamethasone vs. untreated controls.

Teaching Value: This is the canonical dataset for the DESeq2 Bioconductor workflow. It is perfect for teaching paired experimental design and how to control for baseline biological variation (donor-specific effects) using a linear model (`~ cell_line + treatment`).

Link: <https://doi.org/10.1371/journal.pone.0099625>

2. The "Mouse Mammary" Study: Epithelial Populations (Sheridan et al., 2015)

Context: Transcriptomic signatures of three isolated epithelial cell populations from mouse mammary glands.

Teaching Value: The canonical benchmark for the edgeR and limma-voom workflows. It introduces technical batch effects (sequencing lanes) and teaches students how to use PCA/MDS to identify these artifacts and regress them out. It also introduces multi-group contrast matrices.

Link: <https://doi.org/10.1186/s13058-015-0567-z>

3. The "Pasilla" Knockdown Study: Drosophila (Brooks et al., 2011)

Context: Investigating the effect of RNAi knockdown of the pasilla (ps) splicing factor in *Drosophila melanogaster*.

Teaching Value: Another foundational dataset widely used in Bioconductor tutorials. It provides an excellent, simple biological question (knockdown vs. control) with clear signal, ideal for a

student's very first foray into running a differential expression pipeline from start to finish.

Link: <https://doi.org/10.1101/gr.108662.110>

Part 2: Host-Pathogen Interaction Studies (Dual & Complex RNA-seq)

These advanced studies introduce in silico read separation, continuous covariates, and in vivo tissue heterogeneity.

4. Bacteria-Host: The Salmonella Dual RNA-seq Benchmark (Westermann et al., 2016)

Context: Dual RNA-seq of HeLa cells infected with *Salmonella typhimurium*.

Teaching Value: The foundational Dual RNA-seq paper. Students learn to handle mixed-species reads by aligning raw FASTQ data to a concatenated Human+Salmonella reference genome to separate the transcriptomes computationally.

Link: <https://doi.org/10.1038/nature16547>

5. Bacteria-Host: In Vivo Mycobacterium Tuberculosis (Pisu et al., 2020)

Context: Profiling Mtb inside mouse alveolar and interstitial macrophages from living tissue.

Teaching Value: Introduces the challenge of severe read imbalance. Bacterial RNA constitutes a tiny fraction of the pool *in vivo*. It teaches students physical and computational enrichment strategies for low-abundance transcripts.

Link: <https://doi.org/10.1016/j.celrep.2020.107501>

6. Virus-Host: The SARS-CoV-2 Cytokine Signature (Blanco-Melo et al., 2020)

Context: Transcriptomic signature of COVID-19 across cells, animals, and humans.

Teaching Value: Teaches cross-species data integration and the identification of a disease-driving gene expression signature (low interferons, high inflammatory chemokines).

Link: <https://doi.org/10.1016/j.cell.2020.04.026>

7. Virus-Host: Influenza A and Genetic Susceptibility (Alberts et al., 2015)

Context: Host genetic influence on the response to H1N1 Influenza A in mice.

Teaching Value: Viral reads are captured alongside host mRNA. Students learn to quantify these unmapped (viral) reads and use them as a continuous "inferred viral load" covariate in linear models.

Link: <https://doi.org/10.1186/s12864-015-1323-9>

Comprehensive Summary Table

Study Category	Author / Year	Subject / System	Key Teaching Concept	GEO/ BioProject
Foundational	Himes et al., 2014	Human Smooth Muscle (Airway)	Paired design, controlling patient variance (DESeq2)	GSE52778
Foundational	Sheridan et al., 2015	Mouse Mammary Epithelial	Batch effects, multi-group contrasts (edgeR/limma)	GSE63310
Foundational	Brooks et al., 2011	Drosophila (Pasilla knockdown)	Basic differential expression (A/B testing baseline)	GSE18508
Host-Pathogen (Dual)	Westermann et al., 2016	Salmonella in Human HeLa	Mixed-species read mapping, non-coding RNA	GSE73225
Host-Pathogen (In Vivo)	Pisu et al., 2020	M. tuberculosis in Macrophages	In vivo heterogeneity, low-abundance read enrichment	GSE144704
Virus-Host (Signature)	Blanco-Melo et al., 2020	SARS-CoV-2 (COVID-19 models)	Cross-species integration, signature definition	GSE147507
Virus-Host (Covariate)	Alberts et al., 2015	Influenza A in Mice	Viral read quantification as continuous covariate	GSE48466

Standard Upstream Pipeline: Raw Reads to Feature Counts

To prepare the data for downstream analysis in DESeq2 and visualization in ggplot2, students must process raw FASTQ files into a raw count matrix. This generalized pipeline accommodates both standard and mixed-species (Dual) RNA-seq workflows.

Step 1

Quality Control (QC)

Assess raw reads for quality, adapters, and GC content.

```
fastqc *.fastq.gz -t 4 -o fastqc_out/  
multiqc fastqc_out/ -o multiqc_out/
```

Step 2

Read Trimming

Remove low-quality bases and sequencing adapters using Trimmomatic.

```
trimmomatic PE -threads 4 raw_R1.fastq.gz raw_R2.fastq.gz trimmed_R1.fastq.gz  
unpaired_R1.fastq.gz trimmed_R2.fastq.gz unpaired_R2.fastq.gz  
ILLUMINACLIP:TruSeq3-PE.fa:2:30:10 LEADING:3 TRAILING:3 MINLEN:36
```

Step 3

Reference Genome Preparation

For Dual RNA-seq, concatenate host and pathogen genomes. For foundational studies, use a single reference (e.g., Human GRCh38).

```
cat human_genome.fa salmonella_genome.fa > combined_genome.fa  
cat human_annot.gtf salmonella_annot.gtf > combined_annot.gtf  
STAR --runMode genomeGenerate --genomeDir star_index/ --genomeFastaFiles  
combined_genome.fa --sjdbGTFfile combined_annot.gtf
```

Step 4

Splice-Aware Alignment

Map reads to the reference index using STAR.

```
STAR --runThreadN 8 --genomeDir star_index/ --readFilesIn trimmed_R1.fastq.gz  
trimmed_R2.fastq.gz --readFilesCommand zcat --outSAMtype BAM SortedByCoordinate  
--outFileNamePrefix sample_01_
```

Step 5

Quantification (Feature Counting)

Count reads overlapping annotated genes using featureCounts to generate the raw count matrix for R/DESeq2.

```
featureCounts -T 8 -p -a combined_annot.gtf -o raw_counts.txt  
sample_01_Aligned.sortedByCoord.out.bam sample_02_Aligned.sortedByCoord.out.bam
```